

On the quantitative Manin–Mumford conjecture

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Motivation: The Manin–Mumford Conjecture

- K field of characteristic 0,
- X smooth projective curve of genus $g \geq 2$ defined over K ,
- $D \in \text{Div}^1(X)$ degree 1 divisor on X ,
- $j_D: X \rightarrow J_X = \text{Pic}^0(X)$, $P \mapsto P - D$ Abel–Jacobi map.

Manin–Mumford Conjecture (Raynaud’s Theorem 1983)

$$\#j_D(X(\overline{K})) \cap J_X(\overline{K})_{\text{tors}} < \infty.$$

Further Questions:

- a Positive Characteristic:** Naive analogue is false: If $\overline{K} = \overline{\mathbb{F}_p}$, every $\overline{K}$ -point is torsion. Scanlon (2001) and Pink–Roessler (2004) found and proved a suitable analogue.
- b Uniformity:** Is $\#j_D(X(\overline{K})) \cap J_X(\overline{K})_{\text{tors}}$ bounded uniformly in g ?
Proved by Kühne (2021) in characteristic 0.
- c Quantitative Bounds:** For example,
 $\#j_D(X(\overline{K})) \cap J_X(\overline{K})_{\text{tors}} \leq p^{4g} 3^g (p(2g - 2) + 6g)g!$
by Buium (1996) for number fields K , where p is a prime depending on the reduction behavior of X and K .

Quantitative Manin–Mumford-type result

In this talk we want to address all three questions in the case of curves not definable over number fields or finite fields.

Theorem 1 (Looper–Silverman–W.)

K a field, $\mathbb{F}$ its prime field (i.e. $\mathbb{Q}$ or $\mathbb{F}_p$), and X/K a smooth projective geometrically connected curve of genus $g \geq 2$. Assume that there does not exist a finite field extension $\mathbb{F} \subseteq \mathbb{F}'$ and a curve $C/\mathbb{F}'$ such that $C \otimes_{\mathbb{F}'} \bar{K} \cong X \otimes_K \bar{K}$. Then for any $D \in \text{Div}^1(X)$ it holds

$$\#j_D(X(\bar{K})) \cap J_X(\bar{K})_{\text{tors}} \leq 16g^2 + 32g + 124.$$

The function fields case

Theorem 1 will follow from the following special version of non-isotrivial curves over function fields.

Theorem 2 (Looper–Silverman–W.)

B/k smooth projective curve, $K = k(B)$ its function field, $g \geq 2$. For any smooth projective geometrically connected non-isotrivial curve X/K of genus g and any $D \in \text{Div}^1(X)$ it holds

$$\#j_D(X(\overline{K})) \cap J_X(\overline{K})_{\text{tors}} \leq 16g^2 + 32g + 124.$$

Remark: A Bogomolov-type result

We also proved a Bogomolov-type version of Theorem 2. If

$\epsilon \in \left[0, \max\left(\frac{1}{8g}, \frac{1}{9(g-1)}\right)\right)$, then

$$\#\left\{P \in X(\overline{K}) \mid h_{\text{NT}}(j_D(P)) \leq \epsilon \omega^2\right\} \leq \left\lfloor \frac{16g^4 + 36g^2 - 26g - 2}{(g-1)^2(1 - \max\{9(g-1), 8g\}\epsilon)} \right\rfloor + 1,$$

where ω^2 is the self-pairing of the admissible dualizing sheaf ω on X .

Theorem 2 $\Rightarrow$ Theorem 1

Theorem 1 (Looper–Silverman–W.)

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$$\#j_D(X(\bar{K})) \cap J_X(\bar{K})_{\text{tors}} \leq 16g^2 + 32g + 124.$$

- We may assume $K = \bar{K}$. Let $K_0 \subseteq K$ be a finitely generated extension of $\mathbb{F}$ such that $X = X_0 \otimes_{K_0} K$ for X_0/K_0 with $\text{trdeg}(K_0/\mathbb{F})$ being minimal. By assumption $\text{trdeg}(K_0/\mathbb{F}) \geq 1$.
- Choose $k_0 \subseteq K_0$ with $\text{trdeg}(K_0/k_0) = 1$ and k its algebraic closure in K .
- $k \cdot K_0 \subseteq K$ is the function field of a smooth projective curve B/k .
- Let now $j_D(P_1), \dots, j_D(P_s) \subseteq j_D(X(\bar{K})) \cap J_X(\bar{K})_{\text{tors}}$ be pairwise different.
- By a specialization argument we find pairwise different points $j_{D'}(P'_1), \dots, j_{D'}(P'_s) \subseteq j_{D'}(X_0(\bar{k}(B))) \cap J_{X_0}(\bar{k}(B))_{\text{tors}}$.
- Apply Theorem 2 to deduce $s \leq 16g^2 + 32g + 124$.

Structure and History of the proof of Theorem 2

First, Looper–Silverman (LS) and I (W) independently proved a weaker version of the theorem. Both proofs can be structured as follows:

A. Bound of Zhang's admissible pairing for torsion points

Bound of the admissible pairing of the torsion points from below in terms of the Green functions of the reduction graphs and from above in terms of $-\omega^2$. The bound in (W) improves the bound in (LS) by factor 7.

B. Bound of Green functions on metrized reduction graphs

By potential theory on metrized graphs one bounds the sup of the Green function of any metrized graph in terms of Zhang's φ -invariant for metrized graphs. The bound in (LS) improves the bound in (W) by factor 7.

C. Lower bound for ω^2 in terms of the φ -invariants

Known before for $\text{char } K = 0$. For $\text{char } K > 0$ only in (W) using the arithmetic Hodge index theorem for adelic line bundles on X^2 .

We got the theorem by merging our work.

Preliminaries: Zhang's admissible pairing

$\text{Pic}_a(X_{\overline{K}})$: Group of admissible adelic metrized line bundles on $X_{\overline{K}}$.

Zhang defined a pairing

$$(\cdot, \cdot): \text{Pic}_a(X_{\overline{K}}) \times \text{Pic}_a(X_{\overline{K}}) \rightarrow \mathbb{R}.$$

- There is a *canonical* admissible metric on the dualizing bundle ω .
- Any $D \in \text{Div}(X_{\overline{K}})$ induces canonically an admissible metric on $\mathcal{O}_{X_{\overline{K}}}(D)$.

$\Rightarrow$ Expressions like $(n_1\omega + D_1, n_2\omega + D_2)$ make sense.

Some Properties:

- ① $(\cdot, \cdot)$ is bilinear and symmetric.
- ② Adjunction formula: $(\omega, P) = -(P, P)$ for any $P \in X(\overline{K})$.
- ③ If $\overline{\mathcal{L}} \in \text{Pic}_a(X_{\overline{K}})$ and $\deg \mathcal{L} = 0$, then $h_{\text{NT}}(\mathcal{L}) = -(\overline{\mathcal{L}}, \overline{\mathcal{L}})$.

- ④ For different $P, Q \in X(K)$:

$$(P, Q) = i(P, Q) + \sum_{v \in |B|} g_v(R_v(P), R_v(Q)), \text{ where}$$

- $\mathcal{X}$ regular model of X over B (assume semistable reduction).
- $i(P, Q)$ intersection number of the closures of P and Q in $\mathcal{X}$.
- $\Gamma_v(X)$ metrized reduction graph of $\mathcal{X}$ at $v \in |B|$.
- $R_v: X(K) \rightarrow \Gamma_v(X)$ restriction map.
- g_v (canonical) Green function on $\Gamma_v(X)$.

Proof A1: Bound of admissible pairings by ω^2

Let

$P_1, \dots, P_s \in X(\overline{K})$ with $j_D(P_i) \in J_X(\overline{K})_{\text{tors}}$ and $F = P_1 + \dots + P_s \in \text{Div}(X)$.

We have to bound s and may assume $s \geq 2$.

As $\deg(s\omega - (2g - 2)F) = 0$, we have

$$0 \leq h_{\text{NT}}(s\omega - (2g - 2)F) = -(s\omega - (2g - 2)F, s\omega - (2g - 2)F).$$

By bilinearity and the adjunction formula $(\omega, P_j) = -(P_j, P_j)$ this implies

$$\omega^2 \leq -\frac{4(g-1)}{s} \sum (P_j, P_j) - \frac{4(g-1)^2}{s^2} \sum (P_j, P_j) - \frac{4(g-1)^2}{s^2} \sum_{j \neq k} (P_j, P_k).$$

Since $P_j - P_k = j_D(P_j) - j_D(P_k)$ is also torsion,

$$0 = h_{\text{NT}}(P_j - P_k) = -(P_j - P_k, P_j - P_k) = 2(P_j, P_k) - (P_j, P_j) - (P_k, P_k).$$

Summing over all $j \neq k$ we get $\sum (P_j, P_j) = \frac{1}{s-1} \sum_{j \neq k} (P_j, P_k)$ and hence,

$$\omega^2 \leq -\frac{4g(g-1)}{s(s-1)} \sum_{j \neq k} (P_j, P_k).$$

A2: Bound of admissible pairings by Green function

From the previous slide:

$$\omega^2 \leq -\frac{4g(g-1)}{s(s-1)} \sum_{j \neq k} (P_j, P_k). \quad (1)$$

We may assume $P_1, \dots, P_s \in X(K)$ after a finite field extension.
For $P_j \neq P_k$ the pairing (P_j, P_k) can be computed by

$$(P_j, P_k) = i(P_j, P_k) + \sum_{v \in |B|} g_v(R_v(P_j), R_v(P_k)) \geq \sum_{v \in |B|} g_v(R_v(P_j), R_v(P_k)).$$

By Elkies' lower bound for metrized graphs by Baker–Rumely

$$\begin{aligned} \sum_{j \neq k} g_v(R_v(P_j), R_v(P_k)) &\geq -s \cdot \sup_{x \in \Gamma_v(X)} g_v(x, x). \\ \Rightarrow \sum_{j \neq k} (P_j, P_k) &\geq -s \cdot \sum_{v \in |B|} \sup_{x \in \Gamma_v(X)} g_v(x, x). \end{aligned}$$

Combining with (1),

$$\omega^2 \leq \frac{4g(g-1)}{s-1} \sum_{v \in |B|} \sup_{x \in \Gamma_v(X)} g_v(x, x). \quad (2)$$

B & C: Bounding Green functions and ω^2 by φ

By potential theory on metrized graphs one bounds the Green function

$$\sup_{x,y \in \Gamma_v(X)} g_v(x,y) \leq \frac{8g^4+18g^2-13g-1}{2g(2g+1)(g-1)^2} \varphi(\Gamma_v(X)), \quad (3)$$

where $\varphi(\Gamma_v(X))$ is Zhang's φ -invariant of the metrized graph $\Gamma_v(X)$.
By the Hodge index theorem for adelic line bundles on X^2 one proves

$$\omega^2 \geq \frac{g-1}{2g+1} \sum_{v \in |B|} \varphi(\Gamma_v(X)). \quad (4)$$

Combining (2), (3) and (4)

$$\frac{g-1}{2g+1} \sum_{v \in |B|} \varphi(\Gamma_v(X)) \leq \omega^2 \leq \frac{4g(g-1)}{s-1} \cdot \frac{8g^4+18g^2-13g-1}{2g(2g+1)(g-1)^2} \sum_{v \in |B|} \varphi(\Gamma_v(X)).$$

As $\omega^2 > 0$, solving for s yields

$$s \leq \left\lfloor \frac{16g^4+37g^2-28g-1}{(g-1)^2} \right\rfloor \leq 16g^2 + 32g + 124,$$

or $s \leq 1$ if $\sum_{v \in |B|} \varphi(\Gamma_v(X)) = 0$
($\Leftrightarrow$ if X has everywhere good reduction).

The number field case

- The same strategy also works when K is a number field and X/K a smooth projective curve of genus $g \geq 2$, except that for the archimedean places no explicit version of Elkies' bound for the Green function is known.
- The archimedean counterpart of the Green function on the reduction graph is given as follows:
 - Let $\sigma: K \rightarrow \mathbb{C}$ be an archimedean place,
 - X_σ the associated Riemann surfaces,
 - $\omega_1, \dots, \omega_g \in H^0(X_\sigma, \Omega_{X_\sigma})$ an ON-basis (w.r.t. $\langle \omega, \omega' \rangle = \frac{i}{2} \int_{X_\sigma} \omega \wedge \bar{\omega}'$),
 - $\mu = \frac{i}{2g} \sum_{j=1}^g \omega_j \wedge \bar{\omega}_j$,
 - $G^2: X_\sigma^2 \rightarrow \mathbb{R}_{\geq 0}$ the unique function with $\frac{\partial_P \bar{\partial}_P}{2\pi i} \log G^2(P, Q) = \mu(P) - \delta_Q(P)$ and $\int_{X_\sigma} \log G^2(P, Q) \mu(P) = 0$.
- Elkies proved that for pairwise different points $P_1, \dots, P_n \in X_\sigma$ it holds

$$\sum_{j < k} \log G(P_j, P_k) \leq \frac{n \log n}{8\pi} + c(X_\sigma)n$$

for some non-explicit constant $c(X_\sigma)$.

Explicit Elkies bound I

Theorem 3 (W.)

X a compact and connected Riemann surface of genus $g \geq 1$. $\{z_j: U_j \rightarrow \mathbb{C}\}_{j \in J}$ a family of local coordinate functions on X . Assume there exist $r_2 > r_1 > 0$ with $r_2 - r_1 \leq 1$ such that $X = \bigcup_{j=1}^m z_j^{-1}(\mathbb{D}_{r_1})$ and $z_j(U_j) = \mathbb{D}_{r_2}$ for all $j \in \{1, \dots, m\}$. We set

$$C_0 = \sup_{j \in J} \sup_{P, Q \in U_j, P \neq Q} \left| \log |z_j(P) - z_j(Q)| - g(P, Q) \right|,$$

where $g(P, Q) = \log G(P, Q)$. Choose $C_1 > 0, 0 < C_2 \leq \frac{e^{4C_0}}{2\pi}$ such that

$$C_2 \operatorname{id} z_j d\bar{z}_j \leq \mu \leq C_1 \operatorname{id} z_j d\bar{z}_j \text{ on every } U_j$$

For any $n \geq 1$ and any pairwise different points $x_1, \dots, x_n \in X$ it holds

$$\begin{aligned} \sum_{j < k}^n g(x_j, x_k) \leq & n \left(\left(\frac{C_1 e^{4C_0}}{2C_2} + 1 \right) (\log n + 1) + \frac{3}{2} g \log g + 3 - \frac{g+1}{g} H(X) \right) \\ & + nC_0 - n \log(r_2 - r_1). \end{aligned}$$

Explicit Elkies bound II

Theorem 3 (W.)

(...) For any $n \geq 1$ and any pairwise different points $x_1, \dots, x_n \in X$ it holds

$$\sum_{j < k}^n g(x_j, x_k) \leq n \left(\left(\frac{C_1 e^{4C_0}}{2C_2} + 1 \right) (\log n + 1) + \frac{3}{2}g \log g + 3 - \frac{g+1}{g} H(X) \right) + nC_0 - n \log(r_2 - r_1).$$

- Here, $-H(X) = -\int_X \log \|\theta\| \frac{\nu^g}{g!}$ can be bounded by the Faltings height.
- Bounding Elkies' constant $c(X_\sigma)$ in terms of the Arakelov self-intersection number ω^2 would allow to adapt the proof of Theorem 2 to the number field case.
- Theorem 3 is not enough for such a translation, but there is hope that a quantitative Manin-Mumford-type result can be deduced for certain classes of curves. For example, the constants $r_1, r_2, C_0, C_1, H(X)$ has been computed for the modular curves $X_1(pl)$ in terms of pl for two different primes p, l by Edixhoven and de Jong.

Thank you for your attention!